

Candidate Number _____ Fraction _____

2021 32nd Republic of China Biology Olympiad National Team Selection**Camp: Animal Anatomy, Systematics and Ecology****Practical Training A**

All equipment and materials required for this examination have been placed on the table. Please carefully check the items listed on the checklist. If anything is missing, please raise your hand immediately to report it to the examiner.

I. Instructions for Experimental Apparatus:

Equipment 1 (Please leave it on the table during scene changes)		Equipment 2 (one set per person, no reuse)	
name	quantity	name	quantity
Duplex microscope,	2 insect pins (for fixing organs, stored in scintillation counting bottles) 2 pins (2 each		8, 10,
pointed tweezers,	of blue, green, red, white, and black) 1 large dissecting tray 1 small		1, 1, 2,
pointed scissors,	dissecting tray 1 packet of		2
small wrench	20% sodium hydroxide		tubes,
tissues	(NaOH) A and B 1 strip		2
Cleaning	Screw A (blue screw; contained in a 50 mL centrifuge tube) 1 tube		pieces
cloth, oil-based pen,	B snail (green snail; placed in a 50 mL centrifuge tube), 1 bottle of		
timer, dropper,	distilled water, 2 glass		1 bottle,
sealing	slides, 1 can of		2
glue, cover	toothpicks, 1 box.		tablets, 2 tubes
glass slide			
		Students with abnormal color perception should immediately raise their hands to inform the invigilator so that they can be replaced with "text markers".	

Important Notes

- The total time allotted for this exam, including Practical Exercise A and Practical Exercise B, is 90 minutes.
- Please ensure that your exam number on each page is correct. If there is an error, please raise your hand and ask the teaching assistant for assistance.
- Materials, medicines, equipment, and anatomical samples on the table will not be replenished after use.
- Some exam questions will require on-site grading by the invigilator; please be sure to manage your time effectively.
- Please be mindful of safety. In case of an accident, you must notify the invigilator present for assistance.
- This exam paper (including the cover and question paper) consists of 4 pages and must be returned in its entirety upon submission.
- Please answer the questions on this paper. Answers may be written on the back of the questions, but please indicate the question number.

Experiment Topic: Observation and Comparison of Internal Structures of Gastropoda (Morphozoa)

The phylum Mollusca, derived from the Latin word "molluscus," meaning "soft," refers to soft-bodied invertebrates. The common name for organisms belonging to this phylum is molluscs. The phylum Mollusca encompasses a wide variety of species, including common examples such as snails, slugs, abalone, snails, clams, oysters, mussels, razor clams, cockles, octopuses, and squid. Modern classification of mollusks, based on symmetrical body structure, anatomical features such as nerves, ctenoids, radula, and locomotor organs, as well as differences in shell development and style, divides them into five classes: Bivalvia, Cephalopoda, Gastropoda, Polyplacophora, and Scaphopoda. Mollusks are widely distributed and highly adaptable, exhibiting diverse external morphologies. However, most generally possess a soft, moist, and unsegmented body, which can be divided into four parts: head, foot, trunk (visceral mass), and mantle. The head, with its mouth, is usually located at the anterior end of the body and has well-developed eyes or sensory tentacles. The foot, composed of well-developed muscles, is usually located on the abdomen and primarily functions for locomotion; its development varies among species due to adaptations. The visceral mass contains the main internal organs. The mantle covers the mollusk's trunk, and many species can secrete a hard calcareous shell through glandular cells in the mantle epidermis to protect their soft body (modified from: <http://digimuse.nmns.edu.tw/da/collections/az/mo/intro/>).

The radula (plural radulae) is a unique structure found only in mollusks, present in all mollusks except for bivalves. It is a specialized organ used by mollusks to aid in feeding. The radula grows from the radular membrane and contains chains of tiny chitinous structures called "dentals" or "lingual teeth." These structures act like a file, scooping up or cutting food and transporting it directly into the esophagus. Because the diet of mollusks is incredibly diverse, including microalgae, large benthic organisms, attached algae, organic debris, small arthropods, shellfish, and even fish, some mollusks feed on them. To adapt to this diverse diet and ecological environment, the shape, size, and arrangement of the radula vary depending on their diet.

However, there are differences, so the radula is one of the important classification criteria for mollusks.

The experimental materials include two snails of each type, A and B. Please use them to observe, dissect, and compare their internal structures.

Precautions for experimental operation:

1. This question group is divided into two parts. Some questions require the examiners to photograph and score them for verification. 2. Part

1 must be scored before proceeding to Part 2. Failure to do so will result in disqualification from Part 1.

Rate it.

3. Each question will be graded only once. The first and second questions will be graded separately. Please be sure to manage your time and confirm your answers.

After completing the task, raise your hand to notify the examiner to give a score. Once the bell rings at the end of the exam time, no further scoring will be given.

4. Please be careful when using dissection instruments. If injured, immediately raise your hand to inform the invigilator for assistance. 5. Students with abnormal color perception should immediately raise their hand to inform the invigilator so that they can be replaced with "letter pins".

First section: Identifying the internal structure of a snail.

1. Using the provided experimental equipment and referring to "Figure 1", take two types of snails, A (blue snail) and B (green snail).

One snail each. On a large dissecting tray, first use a small wrench to break the shell and remove the shell, then dissect the internal organs, and attach...

On the small dissecting tray, divide the viscera into left and right sides, fix the viscera mass with insect pins, and then follow the instructions in the table below to arrange the corresponding colors.

Insert the pins into the designated organs. After completing the labeling, place the technology sponge next to your seat and raise your hand to ask the judge to review it.

The teacher will take photos and grade the test. Please mark your test number and screw number (e.g., test number **A**) on the foam of the small dissection tray with a pen .

(Note: Only one grade will be given) (1 point for each item, 10 points in total)

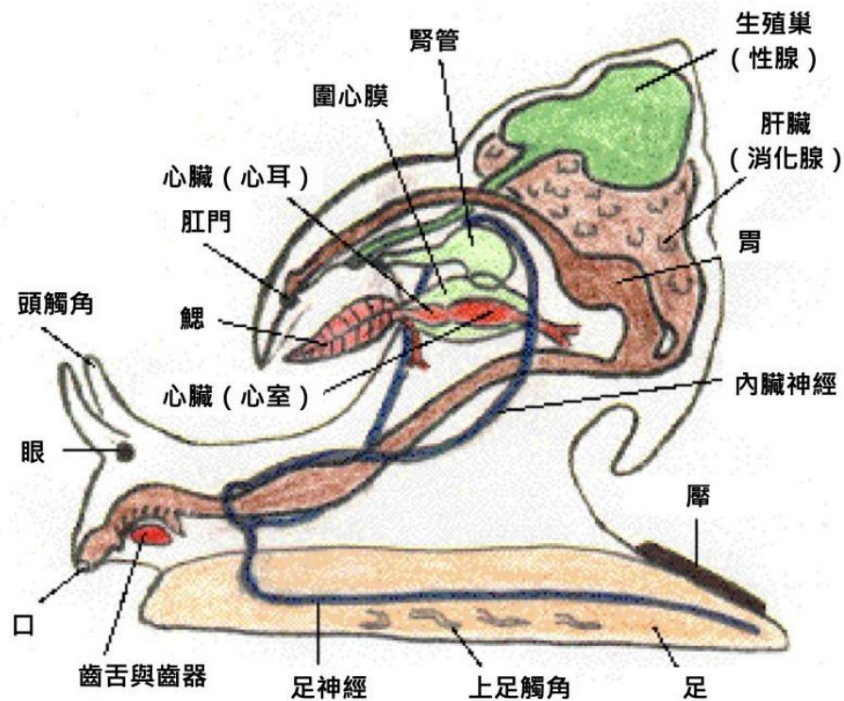


Figure 1. Reference diagram of the anatomy of Gastropoda, Phylum Mollusca.

Indicator needle color	Construct name
blue	liver
green	heart
red	reproductive nest
White	gill
black	Stomach

Please note! You must complete the first section of the question and score it before you can proceed to the second section!

Second section: Anatomy and comparison of the dental and lingual structures.

2. Remove the radulas of both snails , A (blue snail) and B (green snail), and place them separately in two tubes of 20% NaOH solution (A and B) for five minutes. Then rinse with distilled water to remove any remaining tissue. After completion, place the two radulas on glass slides, add embedding solution, and cover with coverslips to create permanent specimens. Please indicate the examination number and snail number on the left side of the slide (e.g., examination number A).
3. Place this permanent specimen in the center of the field of view of the two provided microscopes and observe it at **40x** magnification . Draw a scientific diagram below to represent the observed dentinal structure. After completing the diagram, please raise your hand to notify the teaching assistant and show them the dentinal structure observed under the microscope. (Note: Only one grade will be given, and grading is only possible after the diagram is completed; both specimens must be placed in the center of the field of view of the two microscopes, and the slides must be clearly labeled.) (15 points per specimen, 30 points in total) Please draw the dentinal structure you observed in the table below.

A snail (blue snail)	B snail (green snail)

4. Following on from the previous question, after observing the radulas of the two snails, it is known that these two snails have different diets: one is herbivorous and the other is carnivorous. Please deduce which of A (blue snail) and B (green snail) is herbivorous and which is carnivorous? Please explain your reasoning. (10 points)

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2021 32nd Republic of China Biology Olympiad National Team Selection**Camp: Animal Anatomy, Systematics and Ecology****Practical Training B**

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I. Instructions for Experimental Apparatus:

Number of live animals (one group per person, no repetition)		equipment	
name		name	quantity
Swordfish A (Black Swordtail)	1 petri dish, 1 pair of		1
Swordfish B (Red Swordtail)	pointed forceps (do not injure the fish), 1 dropper (do not		piece,
Swordfish C (Black Spotted Shorttail)	share with Experiment A), 1 tissue (share with Experiment		1 stick,
Swordfish D (Red Shorttail)	A), insect needle (for picking out the copulatory		1 pack
	apparatus, share with Experiment A), cotton pad, dissecting microscope,		of 8
	fishnet		sticks,
			1 pack,
			1 unit
Abnormal color perception does not affect answering questions.			

Important Notes

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Experimental Topic: Reproductive Isolation and Interspecific Identification in Poeciliidae Fishes

In animals that undergo internal fertilization, male external genitalia exhibit a high degree of morphological diversity, showing significant differences even between closely related species, and considerable individual variation within a species. Understanding the evolutionary basis of these interspecific and intraspecific variations is a crucial subject for evolutionary biologists. In all the mentioned...

Among the hypotheses that explain the existence of these diverse forms, the sexual selection hypothesis has received the most support from scholars.

Sexual selection hypotheses predict that the evolution of genital size and morphology will respond to reproductive success and mechanical reproductive isolation between species.

While most studies focus on post-mating mechanisms, such as sperm competition and cryptic female selection, a considerable number of studies also indicate that pre-mating sexual selection mechanisms play a significant role.

Swordfish (*Xiphophorus*) and guppies (*Poecilia reticulata*) belong to the Poeciliidae family, a highly diverse genus. Males use a specialized gonopodium in their anal fin to transfer sperm, and this structure exhibits high variation both between and within populations. Furthermore, all swordfish in the genus undergo sex reversal; before sexual maturity, the anal fin is indistinguishable from that of other fish, but after sexual maturity, the male's anal fin transforms into a gonopodium composed of four rays (3, 4a, 4b, 5). Many studies indicate that the length of the gonopodium in many Poeciliidae fish is influenced by pre-mating selection pressures in females, while other studies suggest a link between gonopodium length and forced mating behavior in males. Due to the complex morphology of the gonopodium in Poeciliidae fish, international scholars have long explored various ecological, evolutionary, and developmental issues, including gonopodium development, the influence of environmental factors, intraspecific variation, and interspecific differences.

The purpose of this practical exercise is to teach students how to determine which of the four swordfish (A, B, C, and D) might be of the same species based on the morphology of their breech features.

How many species are there in total? What is the relationship between this and other morphological characteristics?

Precautions for experimental operation:

1. Please read the instructions carefully, especially the structure of the connector, before operating (see Appendix 1). 2. When catching fish,

keep the seam side of the net facing outwards to avoid scratching the mucous membranes with rough fabric. 3. After catching the

fish, quickly wrap it with a damp cotton pad and place it in the culture dish. 4. If the fish jumps out of the culture dish,

do not scream. Simply use the net and a damp cotton pad to carefully wrap it back into the culture dish.

good

5. 5 points will be deducted from the total score if the fish falls to the ground. No points will be awarded if the fish dies or is seriously

injured. 6. There is no need to raise your hand to ask the teacher for a score (as prolonged observation may cause the fish to die). Simply mark the morphology of

the breech organ you observe on the answer sheet. 7. Please

decide for yourself whether to use bottom or top lighting for observation. 8.

If the cotton pad changes color due to covering the fish, this is normal and you should not panic (it is due to carotenoids in the fish).

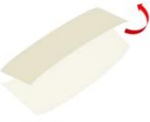
Reusable

9. Because each fish can only be observed for 2 minutes, and each fish can only be picked up and examined once, it is necessary to record the observation immediately.

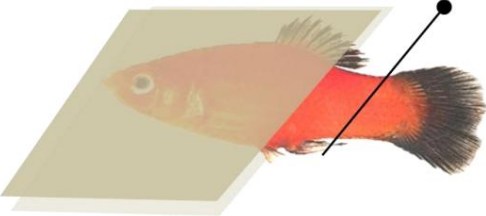
The key points you have grasped

Experimental operation procedure:

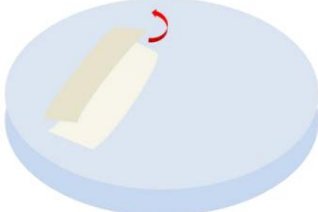
1. 把一片化妝綿一側撕開 (像口袋燒餅一樣)



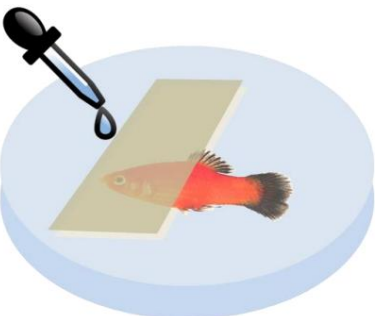
4. 迅速使用昆蟲針小心把臀鰭撥開腹部(不可弄斷或戳傷魚體)以便觀察交接器



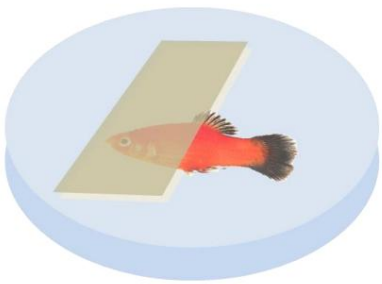
2. 把化妝綿浸水浸濕後擺在培養皿上



5. 觀察期間請使用滴管滴水在化妝綿上以免魚受傷或死亡，但水不能太多讓魚有辦法翻身彈跳。



3. 從魚缸中撈出一條魚後魚頭朝左，以溼的化妝綿迅速包住魚臀鰭之前的身體避免跳動



6. 觀察完交接器形態特徵後馬上把魚放回水中，1-6整個過程不可超過兩分鐘。

After observation, please answer the following questions:

1. Please fill in the correct scientific names and identification characteristics of fish A, B, C, and D. Write O, not X.

(1 point per square, 20 points in total)

	Number	scientific name (just write the species name)	presence of claw	presence of hook	presence of blade	presence of spines
A	X					
B	X					
C	X					
D	X					

2. Are there any two species (A, B, C, and D) that you would classify as the same species but with different coat colors? What explanation would you offer? (10 points)

3. Are there any two pairs in ABCD that you would classify as the same type but with different connectors? What explanation would you offer? (10 points)

4. The differences in the cross-connectives of ABCD corroborate interspecific differences or intraspecific variation? Why? (10 points)